Chemistry • Alkali metals, halogens and transition elements

Group Properties and Elements

Official exercise • Complete solved answers • Group-trend revision

Chapter 9 • Complete official exercise • 10 MCQs + 5 short + 7 constructed + 7 descriptive + 3 investigative

What this chapter is about

• **Group 1 alkali metals** become more reactive down the group because their single outer electron is lost more easily; they become softer and their melting points decrease.

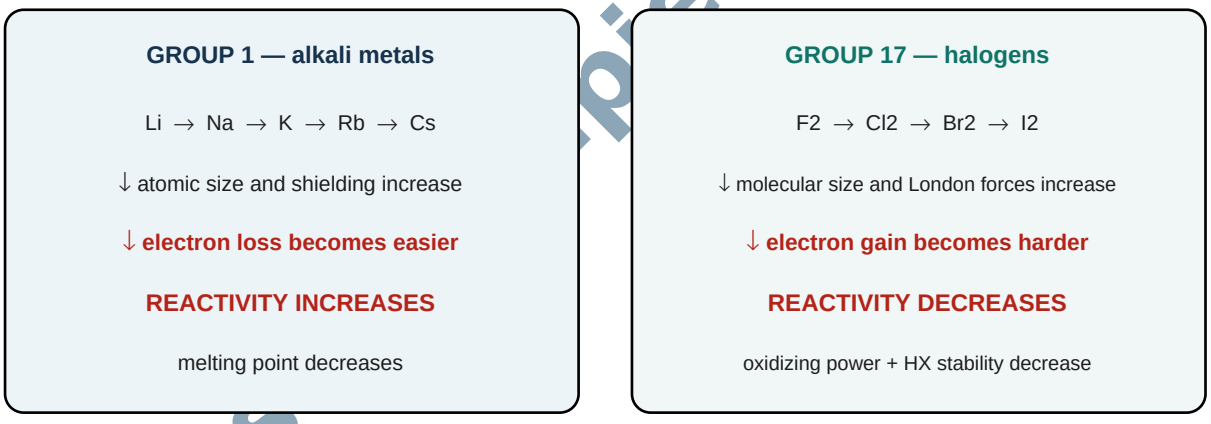
• **Group 17 halogens** become less reactive and weaker oxidizing agents down the group. Their size, density, melting/boiling points and colour depth increase.

• **Transition metals** are dense, high-melting metals with variable oxidation states, coloured compounds and important catalytic uses.

• **Noble gases** are monatomic and very unreactive. Metals and non-metals differ in bonding, structure, conductivity and mechanical behaviour.

Figure 1 — Original comparison of Group 1 and Group 17 trends

Opposite reactivity trends in Groups 1 and 17



Halogen appearance: F2 pale-yellow gas; Cl2 yellow-green gas; Br2 red-brown liquid; I2 grey-black solid.

Metals react by losing electrons; halogens react mainly by gaining electrons.

Exercise coverage: All five official sections are solved in book order. MCQ 4 has no valid printed option; the chemically correct group is stated explicitly.

A. Multiple Choice Questions

1. Which halogen will have the least reactivity with alkaline-earth metals?

(a) chlorine (b) iodine
(c) bromine (d) fluorine

Answer: (b) iodine — Halogen reactivity decreases down group 17, so iodine is the least reactive of the listed halogens.

2. Which compound do you expect to be coloured?

(a) KCl (b) $BaCl_2$
(c) $AlCl_3$ (d) $NiCl_2$

Answer: (d) $NiCl_2$ — $NiCl_2$ contains the transition-metal ion Ni^{2+} and is characteristically green; the other listed salts are white/colourless in ordinary form.

3. In which element do the strongest forces of attraction exist between atoms?

(a) Mg (b) Ca
(c) Sr (d) Ba

Answer: (a) Mg — Metallic bonding in group 2 generally weakens as atoms become larger down the group, so Mg has the strongest attraction among these.

4. Elements of which group are all coloured?

(a) second group (b) sixth group
(c) fourth group (d) fifth group

Answer: No valid option as printed — No listed option is correct. The intended chemical family is **group 17 (historically group 7), the halogens**: F_2, Cl_2, Br_2 and I_2 all have characteristic colours.

5. Which halogen acid is unstable at room temperature?

(a) HBr (b) HI
(c) HCl (d) HF

Answer: (b) HI — HI has the weakest H–X bond in the list and the lowest thermal stability; it decomposes more readily than HBr, HCl or HF.

6. Which oxide is the most basic oxide?

(a) Na_2O (b) Li_2O
(c) MgO (d) CO

Answer: (a) Na_2O — Na_2O is strongly basic and forms NaOH with water. Basic character decreases from alkali-metal oxides towards MgO; CO is neutral.

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7. Which group elements are the most reactive elements?

(a) transition metal group (b) first group
(c) second group (d) third group

Answer: (b) first group — Group 1 alkali metals are the most reactive metal family because they lose their single valence electron very readily.

8. A halogen and a sodium halide solution are mixed. Which mixture turns dark because of a reaction?

(a) Br_2 and NaCl (b) Br_2 and NaF
(c) Cl_2 and NaF (d) Cl_2 and NaI

Answer: (d) Cl_2 and NaI — $Cl_2 + 2NaI \rightarrow 2NaCl + I_2$. Chlorine displaces iodine, whose solution becomes brown/dark.

9. X is a monoatomic gas. Which statement about it is correct?

(a) X burns in air (b) X is coloured
(c) X is unreactive (d) X will displace iodine

Answer: (c) X is unreactive — The description indicates a noble gas: monatomic, colourless and very unreactive due to a complete outer shell.

10. Which property is correct for group 1 elements?

(a) low catalytic activity (b) high density
(c) low electrical conductivity (d) high melting point

Answer: (a) low catalytic activity — Alkali metals are not commonly used as heterogeneous catalysts. They have relatively low densities and melting points and high electrical conductivity.

B. Questions for Short Answers

9.1. Why does it become easier to cut an alkali metal as we move down group 1?

Down the group, atomic size increases and the attraction between metal cations and delocalized electrons becomes weaker. The metallic lattice is therefore easier to deform, so the metals become softer and easier to cut; melting points also decrease.

9.2. Predict the reactivity of potassium towards halogens.

Potassium reacts **very vigorously** with halogens to form ionic potassium halides. Fluorine is most reactive, followed by chlorine, bromine and iodine under comparable conditions.

$$2K(s) + X_2 \rightarrow 2KX(s), \text{ where } X = F, Cl, Br \text{ or } I$$

Example: $2K + Cl_2 \rightarrow 2KCl$

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Potassium is more reactive than sodium and lithium because its outer electron is lost more easily. These reactions are hazardous and are not safe student demonstrations.

9.3. In Cl_2 + 2NaBr(aq) → 2NaCl(aq) + Br_2, chlorine is the oxidizing agent. Which is the reducing agent?

NaBr, specifically Br^-, is the reducing agent. Bromide ions lose electrons and are oxidized: $2Br^- \rightarrow Br_2 + 2e^-$. Chlorine gains those electrons and is reduced to Cl^-.

9.4. Why does iodine exist in the solid state at room temperature?

I_2 molecules are large, contain many electrons and are highly polarizable. Strong London dispersion forces between them require comparatively more energy to overcome, so iodine has a higher melting point and is solid at room temperature.

9.5. How does Ni catalyse the hydrogenation of oil?

Finely divided nickel provides a surface on which hydrogen and unsaturated oil molecules are adsorbed. It weakens/activates the H–H bond and brings the reactants into favourable contact, providing a pathway of lower activation energy.

Hydrogen atoms add across C=C double bonds in the oil. This reduces unsaturation, raises the melting point and converts liquid oil into a more solid fat used for margarine.

$$C{=}C + H_2 \xrightarrow{Ni} C\text{–}C \text{ with two additional C–H bonds}$$

Nickel remains chemically unchanged overall and is separated for reuse; it increases rate but is not consumed as a reactant.

C. Constructed Response Questions

9.1. Which noble gas should have the lowest boiling point and why?

Helium has the lowest boiling point (about −269°C or 4.2 K). It has the smallest atoms and weakest London dispersion forces because only two electrons are present, so very little energy separates its atoms into a gas.

9.2. Compare the reactions of alkali metals with chlorine.

All alkali metals react with chlorine to form white ionic chlorides:

$$2M + Cl_2 \rightarrow 2MCl \text{ (M = Li, Na, K, Rb or Cs)}$$

Reactivity increases down the group as the outer electron becomes easier to lose. Lithium reacts less vigorously, sodium vigorously, potassium violently, and rubidium/caesium extremely violently. The products have M^+ and Cl^- ions in a 1:1 lattice.

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9.3. Why are almost all metals solids while non-metals generally exist as gases and solids?

Most metals have closely packed lattices with strong, non-directional metallic bonding extending throughout the structure. Considerable energy is required to disrupt these lattices, so their melting points are usually high and they are solid at room temperature. Mercury is a notable liquid metal.

Many non-metals form small discrete molecules with weak intermolecular attractions, giving gases such as N_2, O_2 and Cl_2. Others form giant covalent networks or larger molecules, giving solids such as diamond, sulphur, phosphorus and iodine. Bromine is a liquid non-metal.

9.4. Name any three elements in the periodic table which exist as liquids.

Mercury (Hg) and bromine (Br_2) are liquid at ordinary room temperature. **Gallium (Ga)** melts at about 29.8°C and is liquid on a warm day. Caesium (about 28.4°C) is another near-room-temperature liquid; the stated temperature therefore matters.

9.5. Why are transition elements different from normal elements?

Transition elements have partially filled d subshells in at least one common state, while normal/main-group elements have their differentiating electrons in s or p subshells. Their d electrons produce distinctive behaviour:

- Variable oxidation states and often several differently coloured ions/compounds.
- Catalytic activity and formation of complex ions.
- High densities, high melting points, hardness and metallic strength.
- Frequently paramagnetic atoms or ions due to unpaired d electrons.

Main-group elements usually show more direct group-based valence patterns, although they also display important variations.

9.6. Compare the reactivity of chlorine and bromine as oxidizing agents.

Chlorine is the stronger oxidizing agent. Its smaller atoms attract an incoming electron more strongly, whereas bromine's larger, more shielded atoms have a weaker electron-gain tendency.

$$Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$$

$$Br_2 + Cl^- \rightarrow \text{no displacement reaction}$$

Chlorine therefore displaces bromine from bromides, but bromine cannot displace chlorine from chlorides. Oxidizing power follows $F_2 > Cl_2 > Br_2 > I_2$.

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9.7. Which element is the most reactive and which is the least reactive among halogens? Give two reasons.

Among the commonly compared halogens, **fluorine is most reactive and iodine is least reactive**. Astatine is expected to be still less reactive, but its rarity and radioactivity limit ordinary comparison.

1. Atomic size and shielding: Fluorine is smallest, so its nucleus strongly attracts an incoming electron. Down the group, increasing size and shielding weaken that attraction.

2. Electron affinity, electronegativity and bond/oxidizing behaviour: Electron-gain tendency and oxidizing power decrease down the group. Thus fluorine gains electrons most readily, while iodine does so least readily among F, Cl, Br and I.

The same trend explains displacement: a higher halogen displaces a lower halide from solution.

D. Descriptive Questions

9.1. Explain the role of a catalytic converter in an automobile.

A catalytic converter is fitted in the exhaust system. Its honeycomb support provides a large surface area coated with catalysts such as platinum, palladium and rhodium. Hot exhaust gases react on these surfaces, becoming less harmful before release.

Pollutant	Conversion	Purpose
Carbon monoxide, CO	$2CO + O_2 \rightarrow 2CO_2$	Oxidizes toxic CO to CO_2.
Unburned hydrocarbons	Hydrocarbon + $O_2 \rightarrow CO_2 + H_2O$	Completes oxidation of fuel residues.
Nitrogen oxides, NO_x	Example: $2NO \rightarrow N_2 + O_2$	Reduces NO_x towards harmless N_2.

It reduces but does not eliminate pollution: CO_2 is still produced. Unleaded fuel is needed because lead and some other impurities poison the catalyst.

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9.2. Why do chemical reactivities of alkali metals increase down the group whereas they decrease down the group for halogens?

Feature	Group 1 alkali metals	Group 17 halogens
Reaction tendency	Lose one electron to form M^+	Gain one electron to form X^- or attract electrons
Down-group size	Increases	Increases
Shielding and distance	Outer electron is held less strongly	Incoming electron is attracted less strongly
Main energy/property change	Ionization energy decreases	Electron-gain tendency and electronegativity decrease
Result	Reactivity and reducing power increase	Reactivity and oxidizing power decrease

The trends are opposite because metals react mainly by **losing** electrons, while halogens react mainly by **gaining** them.

9.3. Why are metals generally tough and strong whereas non-metals are neither tough nor strong?

Metals have strong, non-directional attraction between cations and delocalized electrons. Layers can shift without completely destroying the bonding, allowing metals to absorb stress, deform and resist fracture.

Many non-metal solids have directional covalent bonds or weak forces between discrete molecules. Applied stress breaks aligned bonds or separates layers, causing brittleness or crumbling; non-metal gases have little structural strength at all.

Exceptions and precision: Diamond is extremely hard because of its strong covalent network, but it is brittle rather than tough. Some non-metal polymers can be flexible and tough, so the statement describes common elemental behaviour, not every material.

9.4. Both alkali metals and halogens are very reactive but have opposite roles. Explain.

Alkali metals have one valence electron and act as **reducing agents**: they lose it to form M^+. Halogens have seven valence electrons and act as **oxidizing agents**: they gain an electron to form X^-.

$$\text{Oxidation: } 2Na \rightarrow 2Na^+ + 2e^-$$

$$\text{Reduction: } Cl_2 + 2e^- \rightarrow 2Cl^-$$

$$\text{Overall: } 2Na + Cl_2 \rightarrow 2NaCl$$

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Their opposite electron tendencies make their mutual reaction vigorous and produce stable ionic salts. Down each group, reactivity increases for alkali metals but decreases for halogens.

9.5. Why is hydrogen bromide thermally unstable compared with hydrogen chloride?

Bromine is larger than chlorine, so the H–Br bond is longer and weaker than the H–Cl bond. Less thermal energy is therefore required to break H–Br into its elements or other products.

Thermal stability: HF > HCl > HBr > HI

Consequently HBr decomposes more readily than HCl on heating. The explanation uses **bond strength**, not the strength of their aqueous acidity.

9.6. Compare the properties of metals and non-metals.

Property	Metals	Non-metals
Usual state at room temperature	Mostly solids; Hg is liquid	Gases, liquids (Br_2) or solids
Lustre	Usually shiny	Usually dull, except examples such as iodine/graphite
Melting/boiling points	Often high	Often low for molecular species; giant networks high
Heat/electricity conduction	Generally good	Generally poor; graphite conducts electricity
Malleability/ductility	Usually malleable and ductile	Solid non-metals usually brittle
Density/strength	Often high and strong	Often lower and mechanically weak or brittle
Electron behaviour	Tend to lose electrons and form cations	Tend to gain or share electrons
Oxides	Usually basic or amphoteric	Usually acidic or neutral

These are general patterns; bonding and structure explain exceptions better than memorizing an absolute rule.

9.7. V_2O_5 catalyst is preferred over platinum in oxidation of sulphur dioxide. Give reasons.

In the Contact process, V_2O_5 catalyses:

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

• **Lower cost:** Vanadium pentoxide is much cheaper than platinum.

• **Poison resistance:** Platinum is readily deactivated by arsenic and other impurities that may occur in SO_2; V_2O_5 is less vulnerable.

• **Durability and supply:** It is robust at the operating temperature and easier to use on a large industrial scale.

Both catalysts increase rate without changing equilibrium yield; the choice is economic and operational, not because platinum is intrinsically unable to catalyse the reaction.

E. Investigative Questions

9.1. Explain the role of sodium as a heat-transfer agent in an atomic nuclear power plant. Which property is utilized?

In a **sodium-cooled fast reactor**, liquid sodium circulates through the reactor core and absorbs heat produced by nuclear fission. It transports this heat to an intermediate heat exchanger; a separate circuit can then produce steam for electricity generation.

• **Excellent thermal conductivity** enables rapid heat transfer.

• **Low melting point (about 98°C) and high boiling point (about 883°C)** provide a wide liquid operating range without requiring very high pressure.

• **Useful heat capacity and pumpability** allow large heat transport through circulation.

Sodium is extremely reactive with water and air, so the coolant circuit must be sealed, oxygen/moisture-free and carefully monitored. Not every nuclear power plant uses sodium; many use ordinary water as coolant.

9.2. Why and how does lithium behave differently from the rest of the alkali metals?

Lithium has the smallest atom and ion in group 1, high charge density and polarizing power, and a relatively high ionization energy. Li^+ is also very strongly hydrated. These factors give lithium compounds more covalent/polarizing character and produce several anomalous behaviours.

• It is harder and less reactive with water than the heavier alkali metals.

• It forms normal oxide Li_2O rather than the peroxides/superoxides common lower in the group.

• It reacts directly with nitrogen to form Li_3N, unlike the other alkali metals.

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• Some lithium salts, including Li_2CO_3, have lower thermal stability or different solubility from their heavier-group analogues.

• It shows a diagonal relationship with magnesium in several chemical properties.

9.3. Why is aluminium used for cooking utensils whereas magnesium is not considered useful?

Aluminium is light, relatively inexpensive, easy to shape and an excellent thermal conductor. A thin, adherent Al_2O_3 layer protects it from further corrosion and makes it practical for pots and pans.

Magnesium is more chemically reactive and less corrosion-resistant in moist or acidic conditions. Magnesium powder, ribbon or thin sections can ignite and burn with an intensely hot flame, making it unsuitable for ordinary cookware. Its reactivity, safety and manufacturing cost outweigh its low density and reasonable thermal conductivity.

The comparison is therefore not based simply on melting point: it depends on conductivity, corrosion protection, reactivity, safety, cost and ease of fabrication.

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Points to Remember

Group 1: reactivity, softness and density generally increase down; melting point and metallic-bond strength decrease.

Group 17: colour deepens and melting/boiling points increase down; reactivity, oxidizing power and hydrogen-halide thermal stability decrease.

A more reactive halogen displaces a less reactive halogen from its halide solution.

In redox, the reducing agent loses electrons; the oxidizing agent gains them.

Transition metals often show variable oxidation states, coloured compounds and catalytic activity.

Noble gases are monatomic and have very low boiling points; helium has the weakest interatomic attractions.

Metals usually conduct and deform because of metallic bonding; non-metal behaviour varies with molecular or network structure.

Catalyst choice depends on activity, cost, lifetime, poison resistance and operating conditions.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	b	2	d	3	a	4	none	5	b
6	a	7	b	8	d	9	c	10	a

How to attempt this chapter in the paper

• For group trends, first state whether the element reacts by losing or gaining electrons, then use size and shielding.
• In displacement equations, compare oxidizing power and identify the species actually oxidized or reduced.
• For thermal stability, discuss H–X bond strength; do not confuse it with aqueous acid strength.
• For catalyst questions, mention lower activation energy and surface adsorption, but do not claim that a catalyst changes equilibrium yield.

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